Math 520, F15 Name: Answer

Quiz 5 SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) Find the general solution of the given differential equation

$$9y'' - 12y' + 4y = 0.$$

Solution: Characteristic equation.

$$9r^2 - 12r + 4 = 0 \Rightarrow (3r-2) = 0 \Rightarrow r_1 = r_2 = \frac{2}{3}$$

Thus the general solution is given by

$$y(t) = c_1 e^{\frac{2}{3}t} + c_2 t e^{\frac{2}{3}t}$$

Problem 2. (6 points) Find the solution of the initial value problem

$$y'' + 4y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = 1.$$

Solution. Characteristic equation.

$$r^2 + 4r + 5 = 0 \Rightarrow r = \frac{-4 \pm \sqrt{4^2 - 4\cdot1\cdot5}}{2} = \frac{-4 \pm \sqrt{4}\,i}{2} = -2 \pm i$$

Thus we get the gereal solution as

$$y(t) = C_1 e^{-2t}\cos t + C_2 e^{-2t}\sin t$$

$$y'(t) = -2C_1 e^{-2t}\cos t - C_1 e^{-2t}\sin t - 2C_2 e^{-2t}\sin t + C_2 e^{-2t}\cos t$$

$$= (-2C_1 + C_2)e^{-2t}\cos t + (-C_1 - 2C_2)e^{-2t}\sin t$$

$$y(0) = 1 \Rightarrow C_1 + C_2 \cdot 0 = 1 \Rightarrow C_1 = 1$$

$$y'(0) = 1 \Rightarrow -2C_1 + C_2 = 1 \Rightarrow C_2 = 1 + 2C_1 = 3$$

So the solution of the IVP is

$$y(t) = e^{-2t}\cos t + 3e^{-2t}\sin t$$